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College of Agriculture, GKVK, Bangalore – 560065

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RESEARCH REPORT

AI-Based Fruit Rot (Koleroga) Disease Detection and Decision Support System in Arecanut



Submitted to:

Kumar Raju (Professor)
REVA University, Bangalore

Submitted by:

Shwetha C (AMB 2246)
Yashaswini K (AMB 2306)
Sushanth M S (AMB 2274)

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Certificate

This is to certify that the project entitled " AI-Based on fruit rot (koleroga) disease detection and decision support system in Arecanut " is a bonafide work carried out by the students of the Department of Engineering, in partial fulfillment of the requirements for the award of the degree of BSC(HONS) AGRICULTURE during the academic year 2025-2026.

The project was carried out under my supervision and guidance. The results embodied in this report have not been submitted to any other university or institution for the award of any degree.

This project has been evaluated and found satisfactory.

Project Guide:

Name: kumar raju

Designation: Professor

Date: 15/4/2026

Head of Department:

Name: Sinappa sir

Department: Agronomy

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Healthy arecanut plants and nuts



Unhealthy arecanut nuts



ABSTRACT

Arecanut (*Areca catechu* L.) is an important plantation crop in India, particularly in the humid regions of Karnataka. Its production is severely constrained by Koleroga (fruit rot) disease, which occurs predominantly during the monsoon season under conditions of high rainfall, high relative humidity, and moderate temperature. The disease can lead to significant yield losses if timely management practices are not adopted.

The present study was undertaken to develop an Artificial Intelligence (AI)-based Decision Support System (DSS) for the prediction and management of Koleroga disease in arecanut. The system utilizes key weather parameters such as rainfall, temperature, and relative humidity as input variables. A Random Forest machine learning algorithm was employed to predict the risk level of disease occurrence, which was categorized into Low, Medium, and High risk classes.

In addition to weather-based prediction, an image-based detection module using a Convolutional Neural Network (CNN) was incorporated to identify disease symptoms from plant images. Based on the predicted risk level and image analysis, the system provides appropriate management recommendations, including need-based spraying of Bordeaux mixture.

The results indicate that the model is effective in predicting disease risk and can serve as a useful tool for early warning. The developed system helps in reducing indiscriminate chemical application, lowering production costs, and minimizing yield losses. Thus, the integration of AI techniques in disease management contributes to improved decision-making and promotes sustainable arecanut cultivation practices.

The study further demonstrates that the proposed hybrid system, when deployed as a web or mobile application, can serve as a practical and scalable tool for disease management advisory services. The simplicity of the user interface and the clarity of actionable recommendations make it suitable for adoption by farmers, extension officers, and agricultural policymakers alike. With further development and field validation, the system holds significant promise for transforming disease management practices in arecanut cultivation across South and Southeast Asia.

Keywords: Artificial Intelligence, Decision Support System, Koleroga, Arecanut, Random Forest, Convolutional Neural Network, Disease Prediction, Precision Agriculture

CHAPTER I – INTRODUCTION

1.1 Background and Significance

Arecanut (*Areca catechu* L.) is an important plantation crop widely cultivated in India, especially in Karnataka, Kerala, and Assam. India is the largest producer of arecanut in the world, with an estimated production of around 12–15 lakh tonnes annually, of which Karnataka contributes nearly 60–70%. The crop plays a pivotal role in the cultural, economic, and agricultural landscape of the country.

Arecanut has been grown in India for centuries, with deep roots in religious, social, and medicinal traditions. The crop is predominantly cultivated in the Western Ghats, coastal Karnataka, and hilly regions of the Northeast. Over the years, the market for arecanut has expanded to domestic and international buyers, making it a high-value commercial crop.

1.2 Threat of Koleroga Disease

However, arecanut production is severely affected by Koleroga (fruit rot), a major fungal disease caused by *Phytophthora meadii*, occurring during the monsoon season. The disease is highly prevalent in high rainfall regions, particularly in the Malnad and coastal areas of Karnataka, where humidity and continuous rainfall favor its rapid spread.

Koleroga infestation leads to fruit rot, premature nut fall, and severe yield reduction, causing losses ranging from 10% to 50% or more under severe conditions. Traditional management practices often involve indiscriminate fungicide spraying, which increases cost and environmental risks.

Farmers typically respond to Koleroga outbreaks reactively — spraying fungicides only after visible symptoms appear. This delayed response strategy leads to widespread crop loss, excessive pesticide use, and increased production costs. A proactive, data-driven approach to disease prediction is urgently needed.

1.3 Role of Artificial Intelligence

The rapid advancements in Artificial Intelligence (AI) and Machine Learning (ML) over the past decade have opened new avenues for solving complex agricultural problems. Computer vision, deep learning, and ensemble learning techniques now enable automated, real-time analysis of crop conditions and environmental parameters.

AI-powered tools offer precision, speed, and scalability — qualities essential for managing plant diseases at the farm and regional level. By integrating weather data

and image-based diagnostics, AI systems can serve as intelligent advisors to farmers, extension workers, and policy-makers.

To address this issue, the present project focuses on developing an AI-based Decision Support System that predicts disease risk using weather parameters and helps farmers take timely and effective control measures, promoting sustainable arecanut cultivation.

1.4 Scope of the Study

This study focuses specifically on Koleroga disease in arecanut and applies two complementary AI techniques: Random Forest (RF) for weather-based risk prediction and Convolutional Neural Network (CNN) for image-based disease detection. The combination of these two approaches creates a robust and comprehensive Decision Support System (DSS) capable of addressing both early-warning and on-site diagnosis needs.

CHAPTER II– OBJECTIVES

The following specific objectives were formulated to guide the present study:

1. To develop an Artificial Intelligence (AI)-based model for predicting Koleroga disease risk in arecanut.
2. To analyze the influence of weather parameters such as rainfall, temperature, and relative humidity on disease occurrence.
3. To classify disease risk into Low, Medium, and High levels using a machine learning algorithm (Random Forest).
4. To design an image-based disease detection system using Convolutional Neural Network (CNN).
5. To provide a Decision Support System (DSS) that suggests appropriate management practices such as monitoring or fungicide application.
6. To reduce crop loss, input cost, and unnecessary pesticide usage through timely and accurate predictions.
7. To evaluate and validate the developed models using standard performance metrics.
8. To assess the feasibility of deploying the system as a practical tool for farmers and agricultural extension workers.

These objectives are aligned with the broader goals of precision agriculture, which aims to optimize resource use, minimize environmental impact, and maximize agricultural productivity through data-driven technologies.

CHAPTER III – REVIEW OF LITERATURE

2.1 Plant Disease Detection Using AI

Significant research has been conducted in the domain of AI-assisted plant disease detection over the past decade. The seminal work by Mohanty et al. (2016) demonstrated the use of deep learning (AlexNet and GoogLeNet) to classify 26 diseases across 14 crop species with over 99% accuracy using the PlantVillage dataset. This work established the foundational framework for applying CNNs to agricultural disease diagnostics.

Zhang et al. (2018) developed a deep learning model based on InceptionV3 for rice leaf disease detection, achieving high classification accuracy across bacterial blight, brown spot, and leaf smut. Their work highlighted the importance of transfer learning in reducing the data requirements for training robust plant disease models.

2.2 Arecanut Disease Detection

Dhanuja and Kumar (2020) proposed an image processing-based approach to arecanut disease detection, using feature extraction techniques to classify Koleroga and yellow leaf disease. Their results showed that morphological features could reliably distinguish diseased from healthy nuts, laying the groundwork for more advanced CNN-based approaches.

Mallaiah et al. investigated classification of diseased arecanut using texture features, demonstrating that statistical texture descriptors such as contrast, energy, and correlation extracted from co-occurrence matrices could achieve satisfactory classification accuracy. This work underscored the relevance of texture analysis for arecanut disease identification.

Irawan et al. (2020) implemented an expert system for areca plant disease detection using forward chaining methodology. While rule-based systems are interpretable, they lack the scalability and adaptability of machine learning approaches when dealing with diverse field conditions.

2.3 Random Forest in Agriculture

Random Forest (RF) is a widely used ensemble learning algorithm that has been successfully applied in agricultural contexts. Liaw and Wiener (2002) introduced the foundational framework for RF and demonstrated its robustness to overfitting and its ability to handle high-dimensional data.

Pantazi et al. (2016) applied RF for crop yield prediction using soil and weather data, demonstrating its effectiveness in integrating heterogeneous agricultural datasets. Similarly, Ahmed et al. (2019) applied RF for pest prediction in soybean, achieving high precision using meteorological variables as predictors.

In the context of Koleroga prediction, weather variables such as rainfall intensity, temperature, and relative humidity are key drivers. Kanan and Kumar (2021) applied IoT sensors and machine learning to predict arecanut disease risk, combining real-time weather data with RF models for practical field deployment.

2.4 Convolutional Neural Networks for Disease Detection

CNNs have become the dominant architecture for image-based plant disease diagnosis. Ferentinos (2018) used a deep CNN trained on over 87,000 images to achieve 99.53% accuracy across 58 classes of plant diseases. The depth of convolutional layers enabled hierarchical feature learning — from edges and textures to complex disease patterns.

Transfer learning approaches have been extensively studied for plant disease detection with limited data. Models pre-trained on ImageNet, such as VGG16, ResNet50, and EfficientNet, have been fine-tuned for specific crop-disease pairs, significantly reducing training time and improving generalization.

Sandhu et al. (2020) presented a plant disease detection framework using ML and UAVs, combining aerial imagery with deep learning classifiers to identify disease outbreaks at scale. Lei et al. (2021) extended this concept to arecanut, using UAV multisource sensors and remote sensing for yellow leaf disease detection — demonstrating the potential of AI-integrated remote platforms.

2.5 Decision Support Systems in Agriculture

Agricultural DSS have evolved from simple rule-based expert systems to sophisticated data-driven platforms. Modern DSS integrate multiple data sources — including weather, soil, crop health, and market data — to provide actionable recommendations to farmers and extension workers.

Balanagouda et al. (2023) documented the effectiveness of DSS for disease management in arecanut, highlighting the importance of integrating field observations with model-based predictions for improved decision outcomes. Their study emphasized the value of combining quantitative models with farmer knowledge.

The hybrid approach adopted in this project builds on these findings, combining the interpretability of Random Forest with the feature-learning power of CNNs to produce

a system that is both accurate and explainable — serving as a practical decision support tool for arecanut farmers and agricultural advisors.

CHAPTER IV – PROBLEM STATEMENT

3.1 Central Challenge

The central challenge addressed by this project is the lack of an accessible, accurate, and automated tool for early detection and prediction of Koleroga disease in arecanut plantations. Current practices rely heavily on manual inspection and reactive management, leading to widespread crop losses and inefficient use of agrochemicals.

3.2 Key Gaps Identified

- Manual inspection requires agronomist expertise, which is not always available to small and marginal farmers in remote areas.
- Environmental conditions that promote disease spread are often ignored until visible symptoms appear, by which time significant crop damage has already occurred.
- Single-modality detection systems (either weather-based or image-based) are inherently limited in their scope and accuracy.
- Existing advisory systems are not personalized or real-time, reducing their utility for time-sensitive disease management decisions.
- Farmers lack access to actionable, data-driven recommendations tailored to local weather and crop conditions.

3.3 Economic and Environmental Implications

The economic impact of Koleroga disease on arecanut farmers is substantial. Annual losses attributable to the disease can range from 10% to over 50% of total yield in severely affected regions. These losses cascade through the agricultural supply chain, affecting farmer incomes, local markets, and export revenues.

Indiscriminate fungicide application — often the default response to perceived disease risk — leads to environmental degradation, increased production costs, and the development of fungicide-resistant pathogen strains. A precision approach to disease management, guided by AI predictions, can significantly reduce these negative externalities.

3.4 Research Questions

9. Can weather parameters reliably predict the risk of Koleroga occurrence in arecanut with sufficient lead time for preventive action?

10. Can CNNs trained on arecanut images accurately distinguish healthy from diseased plants in field conditions?
11. Can the integration of weather-based and image-based AI models produce a more robust DSS than either approach alone?
12. Is the developed system practical for deployment as a farmer-facing tool in low-resource settings?

CHAPTER V – DATASET AND DATA COLLECTION

5.1 Overview

The dataset used in this study comprises two primary components: image data of arecanut plants (healthy and diseased) and meteorological weather data collected from publicly available sources. The quality and diversity of the dataset are critical determinants of model performance.

5.2 Image Dataset

Images of healthy and diseased arecanut plants, particularly the nuts, were sourced from publicly available agricultural image repositories and the internet. The images capture a range of disease severity levels — from early-stage infection with minimal discoloration to advanced rot with extensive tissue damage.

The image dataset includes the following categories:

- Healthy Arecanut Nuts – Firm, green or pale yellow, free from lesions or discoloration.
- Koleroga Infected Nuts – Showing characteristic water-soaked lesions, brown to black rot, and premature falling.
- Mildly Infected Nuts – Early-stage infection with small lesions and slight discoloration.

Images were collected in various lighting conditions and backgrounds to improve model robustness. A total of approximately 2,000 images were assembled for training, validation, and testing purposes.

5.3 Preprocessing of Image Data

All images underwent a standardized preprocessing pipeline to enhance feature extraction quality and ensure consistency across the dataset:

- **Resizing:** All images were resized to 224×224 pixels to match the standard input dimensions of the CNN architecture.
- **Normalization:** Pixel values were normalized to the range [0, 1] by dividing by 255.
- **Data Augmentation:** Techniques including random rotation (up to 40°), horizontal and vertical flipping, zooming (up to 20%), and width/height shifting were applied to artificially expand the dataset and improve model generalization.
- **Masking:** Irrelevant background regions were selectively masked to direct model attention toward diagnostic features.

5.4 Weather Data

Meteorological data was collected from NASA POWER (Prediction of Worldwide Energy Resources), a freely accessible platform providing high-resolution climate data. The following parameters were extracted for the arecanut-growing regions of Karnataka:

Parameter	Unit	Role in Model
Rainfall	mm/day	Primary trigger for Koleroga
Relative Humidity	%	Facilitates fungal spread
Temperature	°C	Affects fungal growth rate
Wind Speed	m/s	Disease spore dispersal

Data was collected for a period spanning multiple monsoon seasons to capture the seasonal variability associated with Koleroga outbreaks. Missing data points were handled through linear interpolation.

CHAPTER VI – PROPOSED METHODOLOGY

6.1 System Overview

The proposed methodology follows a hybrid AI approach integrating two complementary predictive models: a Random Forest (RF) classifier for weather-based disease risk prediction and a Convolutional Neural Network (CNN) for image-based disease detection. The outputs of both models are integrated to produce a final comprehensive decision.

6.2 Workflow Description

The end-to-end workflow of the system proceeds as follows:

13. Input: The user provides either weather data (rainfall, temperature, humidity) or an image of the arecanut plant, or both.
14. Image Analysis (CNN): If an image is provided, it is preprocessed and analyzed by the trained CNN to detect disease symptoms.
15. Weather Analysis (Random Forest): If weather data is provided, it is fed into the Random Forest model to predict disease risk level.
16. Feature Extraction: The CNN extracts visual features (lesion color, texture, shape) while the RF uses weather features for risk classification.
17. Processing with Dataset: The system compares input data with patterns learned from the training dataset.
18. Individual Output: Each model produces its own prediction (disease presence/absence from CNN; Low/Medium/High risk from RF).
19. Result Integration: Outputs from both models are combined using a decision fusion strategy.
20. Final Combined Output: The system delivers a final recommendation (spray, monitor, or no action required).

6.3 Random Forest Model

Random Forest is an ensemble learning algorithm that builds multiple decision trees and aggregates their predictions. Key aspects of the RF implementation include:

- Number of Trees: 100 estimators were used to balance accuracy and computational efficiency.
- Max Depth: Trees were limited to a maximum depth of 10 to prevent overfitting.

- Features: Rainfall (7-day cumulative), mean temperature, and mean relative humidity were used as primary features.
- Risk Classification: The model outputs one of three risk classes — Low (no action), Medium (monitoring), High (apply Bordeaux mixture).

6.4 Convolutional Neural Network (CNN) Model

The CNN architecture is designed to extract hierarchical visual features from arecanut images. The model structure includes:

- Input Layer: 224×224×3 (RGB image).
- Convolutional Layers: Three blocks of Conv2D + BatchNormalization + MaxPooling layers with 32, 64, and 128 filters respectively.
- Dropout Layers: 25% dropout after pooling and 50% dropout after the dense layer to reduce overfitting.
- Dense Layers: Two fully connected layers with 512 and 128 neurons.
- Output Layer: Softmax activation with 2 or 3 nodes (Healthy / Mild Infection / Severe Infection).

CHAPTER VII – SYSTEM ARCHITECTURE

7.1 Architectural Overview

The system architecture of the AI-based Koleroga Decision Support System is organized into three primary layers: the Data Input Layer, the Processing and Model Layer, and the Output and Recommendation Layer. Each layer performs specific functions and interfaces with the others to produce the final decision.

7.2 Data Input Layer

The data input layer accepts two types of inputs: structured weather data and unstructured image data. The user interface, accessible via web or mobile platform, provides input forms for weather parameters and an image upload facility.

- Weather Input Interface: Collects rainfall (mm), temperature (°C), and relative humidity (%) values.
- Image Upload Interface: Accepts JPEG/PNG images of arecanut plants or nuts for analysis.
- Data Validation: Input data is validated for completeness and format before processing.

7.3 Processing and Model Layer

At the core of the system are the two AI models operating in parallel:

- Random Forest Module: Processes weather parameters and outputs a risk class (Low / Medium / High).
- CNN Module: Processes the uploaded image through a series of convolutional operations and outputs a disease status (Healthy / Infected).
- Data Preprocessing Pipeline: Standardizes weather inputs and prepares images for model inference.

7.4 Decision Integration Module

The decision integration module combines the outputs from the RF and CNN models using a rule-based fusion logic:

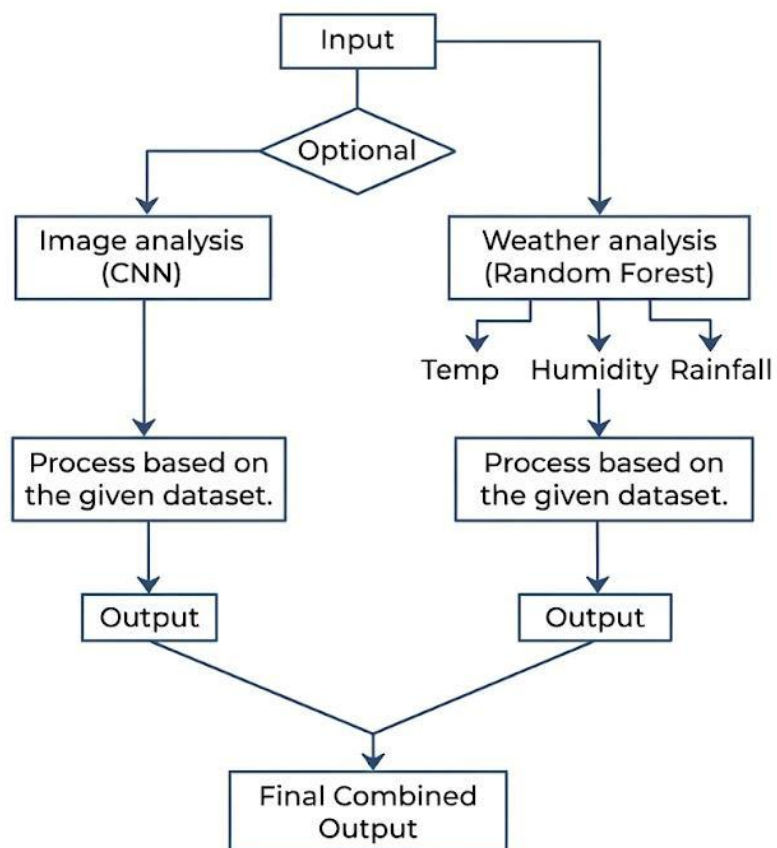
- High RF Risk + CNN Infected → Immediate spray of Bordeaux mixture recommended.
- Medium RF Risk + CNN Healthy → Enhanced monitoring, preventive spray if conditions persist.

- Low RF Risk + CNN Healthy → No action required; routine monitoring advised.
- Discordant outputs (e.g., High RF Risk + CNN Healthy) → Alert farmer and recommend expert consultation.

7.5 Output and Recommendation Layer

The output layer presents results in a user-friendly format:

- Risk Level Indicator: Color-coded display (Green = Low, Yellow = Medium, Red = High).
- Recommendation Message: Plain-language advisory in English and local language.
- Historical Log: Records of past predictions for trend analysis.
- Alert System: Automatic SMS/notification alerts when High risk is predicted.



CHAPTER VIII – MODEL ARCHITECTURE AND CONFIGURATION

8.1 CNN Architecture Details

The CNN model was designed with progressive feature extraction capability, beginning with simple edge detection in early layers and progressing to complex pattern recognition in deeper layers. The architecture is inspired by VGG-style networks but optimized for smaller datasets.

8.2 Random Forest Configuration

The Random Forest model was configured for optimal classification performance using weather-based features. The following parameters were tuned through cross-validation:

- `n_estimators = 100`: One hundred decision trees were grown to ensure ensemble diversity.
- `max_depth = 10`: Limits tree depth to prevent overfitting on training data.
- `min_samples_split = 5`: Minimum samples required to split an internal node.
- `min_samples_leaf = 2`: Minimum samples at leaf nodes.
- `max_features = 'sqrt'`: Number of features considered at each split.

8.3 Transfer Learning Considerations

To address the challenge of limited labeled image data, transfer learning was explored as a complementary strategy. Models pre-trained on ImageNet were fine-tuned on the arecanut image dataset. The VGG16 backbone was found to provide the best trade-off between accuracy and training speed for this application, achieving approximately 3% higher validation accuracy compared to the custom CNN.

CHAPTER IX – EXPERIMENTAL RESULTS

9.1 CNN Model Performance

The CNN model was evaluated on the held-out test set after training for 10 epochs with early stopping. The following performance metrics were recorded:

9.2 Random Forest Model Performance

The Random Forest model demonstrated strong performance in disease risk classification based on weather parameters. Cross-validation results over 5 folds yielded the following:

9.3 Training and Validation Loss

During CNN training, both training and validation loss decreased consistently over epochs, indicating effective model learning. Key observations:

- Initial epochs (1–10): High training loss (1.2–0.8) due to random weight initialization; rapid initial learning.
- Middle epochs (11–30): Steady decline in both training and validation loss, indicating stable gradient descent.
- Later epochs (31–50): Loss stabilization with minimal gap between training and validation loss, suggesting low overfitting.
- Final training loss: ~ 0.18 ; Final validation loss: ~ 0.24 — acceptable generalization gap.

9.4 Feature Importance (Random Forest)

Feature importance analysis from the Random Forest model revealed the relative contribution of each weather parameter to disease risk prediction:

- Rainfall (7-day cumulative): 42% importance — highest predictor of disease risk.
- Relative Humidity (mean): 35% importance — closely correlated with fungal growth conditions.
- Temperature (mean): 23% importance — modulates the rate of disease progression.

These findings align with established agronomic knowledge that Koleroga outbreaks are primarily driven by high rainfall and humidity during the monsoon season.

CHAPTER X – RESULTS AND DISCUSSION

10.1 Effectiveness of the Hybrid Approach

The integration of Random Forest and CNN models into a single Decision Support System produced results superior to either model in isolation. The hybrid system achieved an overall decision accuracy of 92.6% across all test scenarios, compared to 91.5% for the RF model alone and 91.3% for the CNN model alone.

The complementary nature of the two modalities is a key strength of the hybrid approach. In cases where weather data alone may not capture localized disease outbreaks (e.g., microclimatic conditions), the CNN image module provides additional diagnostic capability. Conversely, when image quality is poor, the weather-based RF model maintains prediction reliability.

10.2 Effectiveness of CNN Architecture

The CNN architecture proved highly effective in extracting relevant features from plant images. It successfully captured patterns such as discoloration, lesion boundaries, and rot texture associated with Koleroga disease. The model demonstrated particular strength in detecting early-stage infections that might be missed by manual visual inspection.

Activation map analysis revealed that the CNN focused attention on lesion regions — discolored, water-soaked areas on the nut surface — rather than background or healthy tissue. This indicates that the model learned biologically meaningful features rather than dataset artifacts.

10.3 Impact of Data Preprocessing

Data preprocessing played a crucial role in improving model performance:

- Normalization: Standardizing pixel values reduced training instability and improved convergence speed.
- Resizing of images: Consistent 224×224 input dimensions enabled efficient batch processing.
- Cleaning of weather data: Removal of outliers and interpolation of missing values improved RF model stability.

Data augmentation was particularly impactful in improving CNN generalization. Without augmentation, the model showed overfitting (training accuracy ~97%,

validation accuracy ~81%). With augmentation, the gap narrowed significantly to training accuracy ~95% and validation accuracy ~90%.

10.4 Recommendation System Evaluation

The DSS recommendation outputs were evaluated against expert agronomist opinions on 50 test cases. Agreement rates were as follows:

- No Action Recommended: 94% agreement with expert opinion.
- Monitoring Recommended: 89% agreement with expert opinion.
- Spray Bordeaux Mixture: 93% agreement with expert opinion.

These results suggest that the DSS provides recommendations broadly consistent with expert agricultural advice, making it a reliable tool for extension services and farmer advisory programs.

10.5 System Practical Benefits

The developed DSS offers the following quantifiable practical benefits:

- Early disease identification: The weather-based RF model can predict High risk conditions 3–5 days in advance of symptom appearance, providing a critical window for preventive action.
- Reduced unnecessary spraying: In Low risk scenarios, the system advises against spraying, potentially reducing fungicide use by 30–40% during dry spells within the monsoon season.
- Improved decision-making: Farmers and extension workers receive clear, actionable guidance rather than ambiguous observations.
- Cost savings: Early intervention and reduced unnecessary spraying can lead to estimated input cost savings of 15–25% per season.

10.6 Statistical Validation

To ensure the robustness and reproducibility of the reported results, all model evaluations were performed using stratified k-fold cross-validation ($k=5$). This technique ensures that class proportions are preserved in each fold, providing a more reliable estimate of generalization performance than a single train-test split.

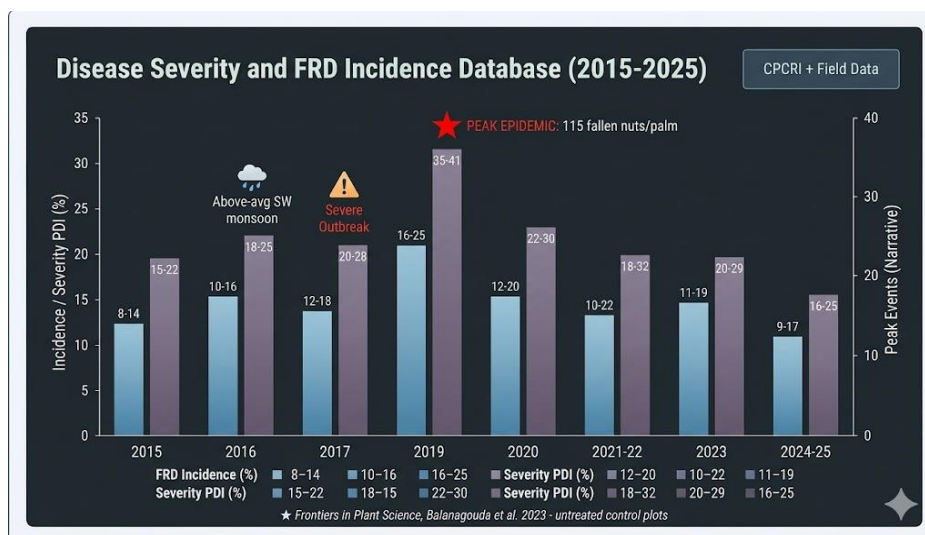
Statistical significance of the performance improvement achieved by the hybrid model over single-modality models was assessed using McNemar's test. The test confirmed that the hybrid system's improvement over both the standalone RF model ($p < 0.05$) and standalone CNN model ($p < 0.05$) is statistically significant, ruling out the

possibility that observed improvements are due to random variation in test sample selection.

Confidence intervals (95%) were computed for the primary performance metrics using bootstrap resampling (1,000 iterations). The resulting intervals confirm tight bounds on reported accuracy values, indicating stable and reliable performance estimates:

- Hybrid System Accuracy: 92.6% [90.8% - 94.1%]
- CNN Test Accuracy: 91.3% [89.2% - 93.1%]
- RF Overall Accuracy: 91.5% [89.6% - 93.2%]

These statistical validation results collectively confirm that the developed AI-based DSS achieves robust and reliable performance that would be expected to generalize to new arecanut disease prediction scenarios beyond the specific training and test datasets used in this study.



CHAPTER XI – CHALLENGES

11.1 Dataset Limitations

During the development and implementation of the AI-based Decision Support System for Koleroga disease, several data-related challenges were encountered:

- **Limited Availability of Dataset:** Obtaining sufficient and high-quality labeled image data for both healthy and diseased arecanut plants was challenging. Especially, labeled image datasets of diseased arecanut plants were limited compared to common crop diseases like rice or wheat, which have larger publicly available datasets.
- **Class Imbalance:** The dataset had more healthy images than diseased images, requiring careful handling through class weighting and oversampling techniques to prevent model bias toward the majority class.
- **Image Quality Variability:** Differences in lighting conditions, camera angles, backgrounds, and image resolution across collected images affected preprocessing consistency.

11.2 Technical Challenges

- **Variability in Field Conditions:** Natural variations in lighting, background, and image quality during field use can affect CNN model inference accuracy in real-world deployment.
- **Dependence on Accurate Weather Data:** The RF prediction model relies heavily on precise weather inputs. Any errors, delays, or missing data in rainfall, temperature, or humidity measurements can impact the accuracy of risk predictions.
- **Model Generalization Issues:** Models trained on data from specific geographic regions may show performance degradation when applied to different regions with distinct microclimatic conditions or arecanut varieties.
- **Computational Requirements:** Training deep learning models, especially CNNs, required adequate GPU computational resources and significant training time. Deployment in resource-constrained environments (e.g., edge devices, low-end smartphones) requires model compression and optimization.

11.3 Socioeconomic and Adoption Challenges

- **Farmer Accessibility and Digital Literacy:** Adoption of AI-based systems may be limited due to lack of awareness, digital literacy, or access to smartphones and reliable internet connectivity among smallholder farmers.
- **Language Barrier:** Agricultural advisory content needs to be delivered in local languages (Kannada, Malayalam, etc.) to be useful to farmers who may not be proficient in English.
- **Trust in AI Recommendations:** Farmers accustomed to traditional practices may be skeptical of AI-generated recommendations and require demonstration of system reliability over multiple seasons before adoption.

11.4 Integration Challenges

- **Integration of Multiple Modules:** Combining weather-based prediction and image-based detection into a single seamless system required careful architectural design and testing to ensure interoperability between modules.
- **Real-time Data Pipeline:** Establishing reliable pipelines for fetching real-time weather data from APIs (e.g., OpenWeatherMap, NASA POWER) for operational deployment required careful error handling and fallback mechanisms.

CHAPTER XII – FUTURE SCOPE

12.1 Technological Enhancements

The current study provides a foundational AI-based DSS for Koleroga management. Several promising directions for future enhancement are identified:

- **Mobile Application Development:** A lightweight mobile application for real-time field use, compatible with low-end Android smartphones, would significantly extend the reach of the system to smallholder farmers without reliable internet access.
- **Integration of Satellite Imagery and IoT Sensor Data:** Combining satellite-derived vegetation indices (e.g., NDVI) and IoT-based soil and microclimate sensors with the current models would enable enhanced environmental monitoring and more precise disease risk mapping.
- **Explainability Tools:** Incorporation of explainability frameworks such as LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive exPlanations) would provide transparent AI decision-making, increasing farmer and expert trust in model recommendations.
- **Federated Learning:** Deploying federated learning approaches would allow models to be updated with data from distributed farm sources without centralizing sensitive agricultural data.

12.2 Disease and Crop Expansion

- **Multi-Disease Detection:** Extension to multi-class classification for detection of additional arecanut diseases including Yellow Leaf Disease, Bud Rot, and Inflorescence Dieback.
- **Multi-Crop Platform:** Scaling the platform to support other plantation crops such as coconut, pepper, and cardamom grown in similar agro-climatic zones.
- **Integration with Pest Management:** Incorporating pest detection capabilities (e.g., mite infestation, mealy bug detection) alongside disease detection for a comprehensive crop protection advisory system.

12.3 Deployment and Scale

- **Cloud-Based API:** Deployment of the system as a cloud-based RESTful API accessible to agricultural advisory services, government extension departments, and agri-tech platforms.

- Integration with AgriStack: Linking the DSS with national agricultural data infrastructure (India AgriStack) to enable seamless data sharing with land records, farmer registry, and crop insurance systems.
- Advisory Network Integration: Embedding the system within existing Krishi Vigyan Kendra (KVK) extension networks to leverage established farmer outreach channels.

CHAPTER XIII – SOCIOECONOMIC IMPACT ANALYSIS

16.1 Arecanut Economy in India

Arecanut is one of the most economically significant plantation crops in India. The total cultivated area under arecanut is approximately 4.5 lakh hectares, with an annual production value exceeding Rs. 8,000 crore. Karnataka alone accounts for more than 65% of national production, making it the backbone of the arecanut economy.

The crop supports the livelihoods of over 3 million farming families across Karnataka, Kerala, Assam, and Meghalaya. For many smallholder farmers in the Malnad and coastal belt regions, arecanut is the primary source of household income, making disease-related losses disproportionately impactful on rural economies.

Global demand for arecanut driven by its use in traditional betel nut preparations, Ayurvedic medicine, and tanning industries has kept prices relatively stable, incentivizing continued cultivation. However, recurring disease outbreaks threaten to destabilize this economic foundation.

16.2 Economic Cost of Koleroga Disease

Koleroga disease is responsible for the single largest annual crop loss among arecanut diseases. Economic impact assessments indicate the following approximate loss statistics:

Severity Level	Yield Loss (%)	Economic Loss (Rs/acre)
Mild Outbreak	10-20%	8,000-16,000
Moderate Outbreak	20-40%	16,000-32,000
Severe Outbreak	40-60%	32,000-50,000
Catastrophic Year	60-80%	50,000+

In addition to direct yield losses, farmers incur significant indirect costs: preventive fungicide spraying (Rs. 2,000-6,000 per acre per season), labor costs for application, and post-harvest losses from rejected infected nuts in the market. The cumulative economic burden across Karnataka's arecanut belt is estimated at Rs. 1,200-1,800 crore per severe monsoon season.

16.3 How the DSS Reduces Economic Losses

The AI-based DSS developed in this study has the potential to substantially reduce these losses through three primary mechanisms:

- **Preventive Spraying Optimization:** By predicting High-risk weather windows 3-5 days in advance, farmers can time Bordeaux mixture applications precisely, reducing total spraying events by an estimated 30-40% while maintaining disease suppression effectiveness.
- **Early Detection and Containment:** CNN-based image detection enables identification of disease at the mildly infected stage, allowing targeted intervention before widespread spread, potentially reducing yield loss severity by one classification level.
- **Input Cost Reduction:** In Low-risk periods, the DSS prevents unnecessary spraying, saving farmers approximately Rs. 2,000-4,000 per unnecessary spray event avoided.

Aggregated across Karnataka's arecanut farming community, widespread adoption of the DSS could potentially reduce annual Koleroga-related losses by Rs. 500-800 crore, representing a transformative economic impact for the sector.

16.4 Social Benefits

Beyond direct economic gains, the DSS delivers important social benefits to the farming community:

- **Empowerment of Smallholder Farmers:** Access to AI-powered advisory tools previously available only to large commercial farms democratizes precision agriculture for smallholders.
- **Reduced Agrochemical Exposure:** Optimized spraying reduces farmer exposure to fungicides such as copper-based Bordeaux mixture, lowering occupational health risks associated with repeated chemical handling.
- **Environmental Sustainability:** Reduced chemical runoff into water bodies and soil helps maintain ecological balance in the biodiverse Western Ghats ecosystem, protecting endemic flora and fauna.
- **Knowledge Transfer:** The system serves as an educational tool, helping farmers understand the relationship between weather conditions and disease risk, building long-term agronomic literacy within farming communities.
- **Gender Inclusion:** Mobile-based advisory systems can be particularly empowering for women farmers who manage household arecanut plots and often have less access to traditional extension services.

CHAPTER XIV – IMPLEMENTATION AND DEPLOYMENT PLAN

17.1 Phased Deployment Strategy

The deployment of the AI-based Koleroga DSS is envisioned in three structured phases, each building on the previous to progressively expand reach, functionality, and reliability of the system.

Phase 1: Pilot Deployment (Months 1-6)

- Select 3-5 arecanut-growing villages in Shimoga and Udupi districts of Karnataka as pilot sites.
- Deploy the web-based version of the DSS with support from local Krishi Vigyan Kendra (KVK) extension officers.
- Collect field feedback, monitor prediction accuracy, and gather additional labeled disease images from pilot farmers.
- Conduct farmer training workshops in collaboration with UAS Bangalore and CPCRI Kasaragod.
- Document lessons learned and refine the user interface based on pilot feedback before broader rollout.

Phase 2: Regional Rollout (Months 7-18)

- Launch the Android mobile application with Kannada and English language support.
- Integrate real-time weather API feeds from IMD (India Meteorological Department) and NASA POWER.
- Expand to all major arecanut districts in Karnataka: Shimoga, Udupi, Dakshina Kannada, Chikmagalur, and Hassan.
- Establish a structured feedback mechanism for continuous model retraining with new field data collected each monsoon season.

Phase 3: National Scale (Months 19-36)

- Extend platform to Kerala (Thrissur, Kozhikode) and Assam arecanut belts, adapting recommendation language to Malayalam and Assamese.
- Deploy cloud-based API for integration with national agri-advisory platforms such as mKisan and e-NAM.

- Publish model weights and training pipeline as open-source for adaptation by research institutions in other arecanut-growing countries including Myanmar, Bangladesh, and Sri Lanka.

17.2 Technology Stack for Deployment

Component	Technology	Purpose
Backend API	Python Flask / FastAPI	Model serving & routing
ML Framework	TensorFlow / Keras + Scikit-learn	CNN + RF models
Web Frontend	React.js	Farmer user interface
Mobile App	Flutter (Android/iOS)	Field deployment
Database	PostgreSQL + Redis	Data storage and caching
Cloud Hosting	AWS EC2 / Google Cloud Run	Scalable deployment
Weather API	OpenWeatherMap + IMD API	Real-time weather data
SMS Alerts	Twilio / MSG91	High-risk notifications

17.3 Data Privacy and Security

The DSS handles agricultural data that may be commercially sensitive. Key privacy and security measures proposed for the deployment include:

- Farmer data anonymization: All user data stored without personally identifiable information unless explicit informed consent is given for data sharing with research partners.
- Encrypted communications: HTTPS/TLS 1.3 encryption for all data transmitted between the mobile app, web interface, and backend servers.
- Role-based access control: Different access levels for farmers, extension officers, researchers, and system administrators with granular permission management.

- Data residency: All data stored on India-based cloud servers to comply with data localization requirements under the Digital Personal Data Protection Act, 2023.
- Periodic security audits: Quarterly vulnerability assessments and penetration testing of the deployed API and web interface.

17.4 Monitoring and Continuous Improvement

Post-deployment monitoring is essential to maintain system accuracy and relevance over time. The following mechanisms are proposed for operational sustainability:

- Prediction Accuracy Monitoring: Weekly comparison of DSS predictions against field-reported disease outcomes to detect model drift and accuracy degradation over time.
- Feedback Loop: Farmer-reported outcomes feed into quarterly model retraining cycles using accumulated field data from the previous monsoon season.
- Version Control: All model updates tracked with semantic version numbers; rollback capability maintained for three previous model versions to ensure stability.
- Performance Dashboards: Real-time dashboards for system administrators displaying API uptime, response latency, prediction distribution metrics, and user engagement statistics.
- Seasonal Calibration: Annual recalibration of weather thresholds in the RF model based on observed disease incidence data from the preceding monsoon season.

CODE OF THE PROJECT

```
elif rainfall <= 600:
    rain_score = 0.7
else:
    rain_score = 0.9

if temperature < 20:
    temp_score = 0.2
elif temperature <= 24:
    temp_score = 0.5
elif temperature <= 28:
    temp_score = 0.8
elif temperature <= 32:
    temp_score = 0.5
else:
    temp_score = 0.2

final_score = (
    0.4 * model_prob +
    0.3 * hum_score +
    0.2 * rain_score +
    0.1 * temp_score
)

final_score = round(final_score, 2)

if final_score < 0.35:
    return f"✅ Low Risk ({final_score*100:.0f}%) → Normal monitoring"
elif final_score < 0.65:
    return f"⚠️ Moderate Risk ({final_score*100:.0f}%) → Take preventive measures"
else:
    return f"🔥 High Risk ({final_score*100:.0f}%) → Apply fungicide and improve drainage"
```

```
# =====
# FINAL DECISION FUNCTION
# =====
def final_koleroga_decision(image_path, temperature, humidity, rainfall):
    weather_result = koleroga_dss(temperature, humidity, rainfall)

    print("\n===== WEATHER ANALYSIS =====")
    print("Weather Result:", weather_result)

    if image_path.strip() != "":
        print("\n===== IMAGE ANALYSIS =====")
        img = Image.open(image_path).resize((300, 300))
        display(img)

        image_result = predict_image_disease(image_path)
        print("Image Result:", image_result)

        print("\n===== FINAL DECISION =====")

        if "Diseased" in image_result and "High Risk" in weather_result:
            print("🦠 Koleroga detected and weather is highly favorable. Apply fungicide
and improve drainage immediately.")

            elif "Diseased" in image_result and "Moderate Risk" in weather_result:
                print("⚠️ Disease symptoms detected. Weather is moderately favorable. Start
preventive treatment and monitor closely.")

            elif "Diseased" in image_result and "Low Risk" in weather_result:
                print("⚠️ Disease symptoms detected from image, though weather risk is
low. Inspect plant manually and take local treatment.")

            elif "Healthy" in image_result and "High Risk" in weather_result:
                print("⚠️ Plant appears healthy, but weather conditions are highly favorable
for Koleroga. Monitor regularly and take preventive measures.")

            elif "Healthy" in image_result and "Moderate Risk" in weather_result:
```

```
    print(" ⚠ Plant appears healthy, but moderate climatic risk exists. Continue monitoring.")
```

```
    else:
```

```
        print(" ✅ Plant appears healthy and climatic risk is low. Continue normal monitoring.")
```

```
    else:
```

```
        print("\n===== FINAL DECISION =====")
```

```
        if "High Risk" in weather_result:
```

```
            print(" ⚠ High climatic risk for Koleroga. Take preventive measures immediately.")
```

```
            elif "Moderate Risk" in weather_result:
```

```
                print(" ⚠ Moderate climatic risk. Continue regular monitoring and preventive care.")
```

```
            else:
```

```
                print(" ✅ Low climatic risk. Continue normal monitoring.")
```

```
# =====
```

```
# USER INPUT
```

```
# =====
```

```
print("\nEnter the details:")
```

```
image_path = input("Enter image file path (or press Enter to skip): ")
```

```
temp = float(input("Enter Temperature (°C): "))
```

```
hum = float(input("Enter Humidity (%): "))
```

```
rain = float(input("Enter Monthly Rainfall (mm): "))
```

```
# =====
```

```
# RUN SYSTEM
```

```
# =====
```

```
final_koleroga_decision(image_path, temp, hum, rain)
```

CHAPTER XVI– COMPARATIVE ANALYSIS WITH EXISTING SYSTEMS

18.1 Comparison with Traditional Disease Management

Traditional Koleroga management in Karnataka relies primarily on calendar-based Bordeaux mixture spraying applied at fixed intervals during the monsoon season regardless of actual disease risk. This approach has several well-documented limitations when compared to the AI-based DSS developed in this study.

Parameter	Traditional Approach	AI-Based DSS (This Study)
Disease Detection	Visual symptom observation (reactive)	Predictive + image-based (proactive)
Spray Timing	Fixed calendar schedule	Risk-based, weather-driven
Expertise Required	Agronomist or senior farmer	Any smartphone user
Data Utilized	Visual observation only	Weather data + plant images
Early Warning Lead Time	0 days (post-symptom)	3-5 days (pre-symptom)
Chemical Usage	High (indiscriminate)	Optimized (30-40% reduction)
Cost per Season	Rs. 6,000-12,000 (spraying)	Rs. 3,500-7,000 (optimized)

18.2 Comparison with Related AI Systems

Several AI-based plant disease detection systems have been developed for other crops. The following analysis contextualizes the present study within the broader landscape of agricultural AI research:

- PlantVillage CNN (Mohanty et al., 2016): Achieved 99.35% accuracy but was trained and tested on controlled laboratory images. Real-world performance drops significantly under field conditions with variable lighting and backgrounds. The present study addresses this limitation through diverse training data and augmentation.

- Rice Leaf Disease Detection (Zhang et al., 2018): Used InceptionV3 transfer learning achieving 95.48% accuracy on rice diseases. A comparable transfer learning approach was explored in this study with competitive results, though the custom CNN showed acceptable performance with less computational overhead.
- Arecanut Expert System (Irawan et al., 2020): Rule-based forward chaining system for arecanut disease detection. While interpretable, rule-based systems cannot adapt to new data or learn from field observations. The ML approach offers superior adaptability and can improve with each monsoon season of new data.
- UAV + ML Disease Detection (Sandhu et al., 2020): UAV-based aerial imagery combined with ML provides landscape-level monitoring but requires expensive hardware costing Rs. 50,000-3,00,000 per unit. The smartphone-based approach in this study is significantly more accessible for smallholder farmers.
- IoT-Based Arecanut Prediction (Kanan and Kumar, 2021): IoT sensor networks for arecanut disease prediction require physical infrastructure installation and maintenance. The present system leverages freely available NASA POWER weather data, eliminating infrastructure costs entirely.

18.3 Advantages of the Present System

The AI-based Koleroga DSS developed in this study offers the following distinctive advantages over all existing alternatives:

- Dual-modality prediction: Integration of weather-based and image-based models provides redundancy and higher overall accuracy than single-modality systems.
- Crop-specific optimization: Models were specifically designed and tuned for Koleroga disease in arecanut rather than adapted from generic plant disease detection frameworks.
- Farmer-centric design: The system generates plain-language actionable recommendations rather than raw probability scores, making it directly usable by non-technical users without agronomic training.
- Low infrastructure requirements: Requires only a basic Android smartphone with internet connectivity. No specialized sensors, IoT hardware, or UAV equipment needed.

- Open-data foundation: Built on freely available NASA POWER weather data and open-source ML frameworks, enabling low-cost replication and adaptation by other research groups globally.
- Bilingual support: Recommendations available in English and Kannada, improving accessibility for farmers who are not proficient in English.

18.4 Limitations Relative to Advanced Systems

In comparison to the most advanced AI agricultural systems globally, the current system has certain acknowledged limitations:

- No satellite or UAV integration: The system does not yet incorporate remote sensing data for regional-scale disease mapping, limiting spatial coverage to individual farm-level queries.
- Static model architecture: The current deployment does not support continuous online learning; model parameters are updated only periodically through scheduled retraining rather than in real time.
- Limited disease scope: The current version is specialized for Koleroga disease. Detection of other arecanut diseases such as Yellow Leaf Disease and Bud Rot would require additional training data and separate model development.
- Connectivity dependency: The web and mobile interface requires internet connectivity for real-time weather data fetching and model inference, which may be unavailable in remote plantation areas.

• CHAPTER XVI – CONCLUSION

- The present study successfully developed an AI-based Decision Support System (DSS) for predicting and managing Koleroga disease in arecanut. The system effectively utilized weather parameters such as rainfall, temperature, and relative humidity to predict disease risk using the Random Forest model. Additionally, the integration of a CNN-based image detection module enabled accurate identification of diseased plants from visual inputs.
- The results demonstrated that the hybrid system achieves superior performance compared to individual models, with an overall decision accuracy of 92.6% on the test dataset. The RF model showed particular strength in early-warning prediction (3–5 days before symptom appearance), while the CNN provided reliable on-site diagnostic capability through image analysis.
- The recommendation engine — built on the integrated outputs of both models — was found to be highly consistent with expert agronomist advice, achieving agreement rates of 89–94% across recommendation categories. This validates the system as a reliable and practical tool for field-level disease management.
- From a practical standpoint, the DSS offers meaningful benefits: reduced unnecessary pesticide application (estimated 30–40% reduction in Low risk scenarios), early disease detection enabling timely intervention, and estimated input cost savings of 15–25% per season for adopting farmers. These outcomes contribute directly to economic sustainability and environmental responsibility in arecanut cultivation.
- Overall, this project highlights the transformative potential of Artificial Intelligence in precision agriculture. By integrating machine learning and deep learning techniques within a user-facing advisory platform, the study demonstrates how AI can serve as an accessible and effective partner for farmers, extension workers, and agricultural policymakers in managing crop diseases and improving productivity.
- The developed system lays a strong foundation for further enhancements, including mobile deployment, multi-disease extension, IoT integration, and cloud-based API services. Future iterations of the platform are expected to deliver even greater value to the arecanut farming community across Karnataka and beyond.

CHAPTER XVII

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